

# Musterlösung Serie 5

## VECTOR SPACES AND SUBSPACES

**1. Linear systems and subspaces.** Let  $m \in \mathbb{R}$ . Describe the solutions of the following system of equations depending on  $m$ :

$$\begin{cases} x + my &= -3 \\ mx + 4y &= 6 \end{cases}$$

When is the set of solutions  $S$  a linear subspace of  $\mathbb{R}^2$ ? Give a geometrical interpretation of  $S$  depending on  $m$ .

*Solution:* We use Gauss elimination. We do not change the first row but we replace the second row  $R_2$  by  $R_2 - mR_1$ . We obtain the equivalent system

$$\begin{cases} x + my &= -3 \\ (4 - m^2)y &= 6 + 3m = 3(m + 2) \end{cases}$$

We now discuss the solutions depending on the value of  $m$ :

- If  $m \notin \{\pm 2\}$ , then  $4 - m^2$  does not vanish. Hence,

$$y = \frac{3(m + 2)}{4 - m^2} = \frac{3(m + 2)}{(2 + m)(2 - m)} = \frac{3}{2 - m}$$

Plugging it into the first equation, we find

$$x = \frac{6}{m - 2}.$$

The system therefore admits a unique solution, namely  $(\frac{-6}{2-m}, \frac{3}{2-m})$ . Geometrically, this means that the two lines defined by the equations  $x + my = -3$  and  $mx + 4y = 6$  intersect in this point.

- If now  $m = 2$ , the last line becomes  $0 = 12$ . Therefore the system doesn't admit any solution. Geometrically, this implies that the two lines are parallel when  $m = 2$ .
- If  $m = -2$ , then the last line becomes  $0 = 0$  and the system is equivalent to  $x = -3 - my$ . We then have

$$S = \{(-3 - my, y) \mid y \in \mathbb{R}\}.$$

Geometrically, this implies that both equations define the same line when  $m = -2$ .

In none of these cases is  $S \subseteq \mathbb{R}^2$  a linear subspace since it never contains  $0 \cdot S = \{(0, 0)\}$ .

**2. Linear subspaces.** Which of the following sets are linear subspaces of the given vector spaces? What changes when  $\mathbb{R}$  is replaced by  $\mathbb{F}_2$  in (b) and (c)?

(a)  $S_1 := \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = x_2 = 2x_3\} \subseteq \mathbb{R}^3$

(b)  $S_2 := \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^4 = 0\} \subseteq \mathbb{R}^2$

(c)  $S_3 := \{(\mu + \lambda, \lambda^2) \in \mathbb{R}^2 \mid \mu, \lambda \in \mathbb{R}\} \subseteq \mathbb{R}^2$

*Solution:*

(a) We see that  $S_1$  is the set of solutions of a homogeneous linear system of equations and thus a linear subspace.

(b) The equation  $x_1^2 + x_2^4 = 0$  has only the solution  $x_1 = x_2 = 0$  in  $\mathbb{R}$ . Therefore, we have  $S_2 = \{(0, 0)\}$  and thus it is a linear subspace.

If we consider the equation over  $\mathbb{F}_2$ , the situation is different. For every  $\lambda \in \mathbb{F}_2$ , we have

$$\lambda^2 = \lambda.$$

Hence, the equation defining  $S_2$  is equivalent to  $x_1 + x_2 = 0$  over  $\mathbb{F}_2$ . As the set of solutions of a homogenous linear equation  $S_2$  is a linear subspace of  $\mathbb{F}_2^2$ .

(c) The set  $S_3$  is not a linear subspace of  $\mathbb{R}^2$ , as for example  $(1, 1)$  is contained in it, but not the multiple

$$(-1) \cdot (1, 1) = (-1, -1),$$

as the square of any real number is positive.

Over  $\mathbb{F}_2$ , we again use the identity  $\lambda^2 = \lambda$ . For any  $x, y \in \mathbb{F}_2$ , we define

$$\lambda = y, \mu = x - y.$$

Then  $(\mu + \lambda, \lambda) = (x, y)$ , hence  $S_3 = \mathbb{F}_2^2$ . In particular, it is also a linear subspace.

**3. Vector spaces of maps.** Let  $K$  be a field in which  $1 + 1 \neq 0$  and consider the space

$$V = K^K = \text{Abb}(K, K) := \{f : K \rightarrow K\}.$$

Recall from the lectures that it is a vector space when endowed with scalar multiplication, namely  $(\alpha \cdot f)(x) = \alpha f(x)$ ,  $\forall \alpha \in K$ ,  $\forall x \in K$ , and with point wise addition, i.e.  $(f + g)(x) = f(x) + g(x)$ ,  $\forall x \in K$ .

Now let

$$V_{\text{even}} := \{f : K \rightarrow K \mid f(-x) = f(x) \forall x \in K\},$$

$$V_{\text{odd}} := \{f : K \rightarrow K \mid f(-x) = -f(x) \forall x \in K\}.$$

Show that  $V_{\text{even}}$  and  $V_{\text{odd}}$  are linear subspaces of  $V$ , that

$$V_{\text{even}} + V_{\text{odd}} := \{v + w \mid v \in V_{\text{even}}, w \in V_{\text{odd}}\} = V$$

and that  $V_{\text{even}} \cap V_{\text{odd}} = \{0\}$ .

*Solution:* First note that the function that vanishes everywhere belongs both subsets, hence they are not empty. Now let  $f, g \in V_{\text{even}}$  and  $a \in K$ . We have

$$\forall x \in K : (f + a \cdot g)(-x) = f(-x) + a \cdot g(-x) = f(x) + a \cdot g(x) = (f + a \cdot g)(x).$$

Hence  $f + a \cdot g \in V_{\text{even}}$  for all  $f, g \in V_{\text{even}}$ , for all  $a \in K$ . This proves that  $V_{\text{even}}$  is a linear subspace of  $V$ .

Similarly, let  $f, g \in V_{\text{odd}}$  and let  $a \in K$ . Then,

$$(f + a \cdot g)(-x) = -f(x) - a \cdot g(x) = -(f + a \cdot g)(x).$$

Hence  $V_{\text{odd}}$  is a linear subspace of  $V$ .

Assume now that  $f \in V_{\text{odd}} \cap V_{\text{even}}$ . Then, for all  $x \in K$

$$-f(x) = f(-x) = f(x),$$

which implies that we must have  $f(x) = 0$  for all  $x \in K$  since  $1 + 1 \neq 0$ .

Finally, we show that  $V_{\text{even}} + V_{\text{odd}} = V$ . Let  $f \in V$  and define

$$f_{\text{even}}(x) := \frac{f(x) + f(-x)}{2}$$

$$f_{\text{odd}}(x) := \frac{f(x) - f(-x)}{2}.$$

You can easily see that  $f_{\text{even}} \in V_{\text{even}}$ , that  $f_{\text{odd}} \in V_{\text{odd}}$  and that

$$f(x) = f_{\text{even}}(x) + f_{\text{odd}}(x).$$

This concludes the proof.

**4. Vector spaces.** Let  $\infty$  and  $-\infty$  denote 2 distinct objects, neither of which is in  $\mathbb{R}$ . Define an addition and a scalar multiplication on  $V := \mathbb{R} \cup \{\infty\} \cup \{-\infty\}$  as follows: in  $\mathbb{R}$ , addition and multiplication are defined as usual. For  $t \in \mathbb{R}$  define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

$$t + \infty = \infty + t = \infty, \quad t + (-\infty) = (-\infty) + t = -\infty.$$

$$\infty + \infty = \infty, \quad (-\infty) + (-\infty) = (-\infty), \quad \infty + (-\infty) = (-\infty) + \infty = 0.$$

Is  $V$  a vector space over  $\mathbb{R}$ ?

*Solutions:* We have  $(1-2) \cdot \infty = -1 \cdot \infty = -\infty$  but  $(1 \cdot \infty) + (-2 \cdot \infty) = \infty + (-\infty) = 0 \neq -\infty$ . So, axiom (V8) doesn't hold and  $V$  isn't a vector space over  $\mathbb{R}$ .

**5. Vector spaces.** Let  $X$  be a set and let  $P$  be its power set (this means that  $P$  is the set of all subsets of  $X$ ). For all  $A, B \in P$  and for  $\lambda \in \mathbb{F}_2$ , define

$$A \triangle B := (A \cup B) \setminus (A \cap B)$$

$$\lambda \cdot A := \begin{cases} \emptyset, & \text{for } \lambda = 0, \\ A, & \text{for } \lambda = 1. \end{cases}$$

Show that  $(P, \triangle, \cdot, \emptyset)$  is a  $\mathbb{F}_2$ -vector space.

*Solution:* Throughout, let  $A, B, C \in P$  and  $s, t \in \mathbb{F}_2$ . We carefully check the axioms.

(V1) We can get intuition for associativity using Venn diagrams. We obtain the following picture:

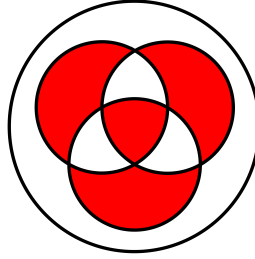


Abbildung 1: Symmetric difference of 3 sets

Let us now prove it. Let  $x \in A \triangle (B \triangle C)$ . We then have

$$\begin{aligned} & x \in A \triangle (B \triangle C) \\ \Leftrightarrow & (x \in A \wedge x \notin B \triangle C) \vee (x \notin A \wedge x \in B \triangle C) \\ \Leftrightarrow & [x \in A \wedge \neg((x \in B \wedge x \notin C) \vee (x \notin B \wedge x \in C))] \\ & \vee [x \notin A \wedge ((x \in B \wedge x \notin C) \vee (x \notin B \wedge x \in C))] \end{aligned}$$

We split the last expression above into the two parts delimited by square brackets and simplify them individually. The second one can be written as

$$(x \notin A \wedge x \in B \wedge x \notin C) \vee (x \notin A \wedge x \notin B \wedge x \in C)$$

by distributing the logical “and ” inside the parentheses.

On the other hand, for the first part

$$\begin{aligned}
& x \in A \wedge \neg((x \in B \wedge x \notin C) \vee (x \notin B \wedge x \in C)) \\
\Leftrightarrow & x \in A \wedge (\neg(x \in B \wedge x \notin C) \wedge \neg(x \notin B \wedge x \in C)) \\
\Leftrightarrow & x \in A \wedge ((x \notin B \vee x \in C) \wedge (x \in B \vee x \notin C)) \\
\Leftrightarrow & x \in A \wedge ((x \notin B \wedge x \in B) \vee (x \notin B \wedge x \notin C) \\
& \quad \vee (x \in C \wedge x \in B) \vee (x \in C \wedge x \notin C)) \\
\Leftrightarrow & x \in A \wedge ((x \notin B \wedge x \notin C) \vee (x \in C \wedge x \in B)) \\
\Leftrightarrow & (x \in A \wedge x \notin B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \in B)
\end{aligned}$$

From line 3 to line 4 as well as from line 6 to line 7 we used the distributive property of the logical “and ” with respect to the logical “or”.

Putting the two parts back together, we have

$$\begin{aligned}
& x \in A \Delta (B \Delta C) \\
\Leftrightarrow & (x \notin A \wedge x \in B \wedge x \notin C) \vee (x \notin A \wedge x \notin B \wedge x \in C) \\
& \vee (x \in A \wedge x \notin B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \in B)
\end{aligned}$$

This last line is the translation into logic from the above Venn diagram.

By following the exact same steps for  $(A \Delta B) \Delta C$ , we obtain exactly the same final expression and conclude that

$$A \Delta (B \Delta C) = (A \Delta B) \Delta C.$$

(V2) We have

$$\emptyset + A = (A \cup \emptyset) \setminus (A \cap \emptyset) = A.$$

(V3) We see that

$$A + A = (A \cup A) \setminus (A \cap A) = A \setminus A = \emptyset.$$

(V4) We see that

$$A + B = (A \cup B) \setminus (A \cap B) = (B \cup A) \setminus (B \cap A) = B + A.$$

(V5) We have

$$\begin{aligned}
(0 \cdot 0) \cdot A &= 0 \cdot A = \emptyset = 0 \cdot \emptyset = 0 \cdot (0 \cdot A) \\
(0 \cdot 1) \cdot A &= 0 \cdot A = \emptyset = 0 \cdot (1 \cdot A) \\
(1 \cdot 1) \cdot A &= 1 \cdot A = A = 1 \cdot (1 \cdot A).
\end{aligned}$$

The rest of the cases are proved by the commutativity of  $\mathbb{F}_2$ .

(V6) It holds by definition.

(V7) We have

$$\begin{aligned} 0 \cdot (A + B) &= \emptyset = \emptyset + \emptyset = 0 \cdot A + 0 \cdot B \\ 1 \cdot (A + B) &= A + B = (1 \cdot A) + (1 \cdot B). \end{aligned}$$

(V8) We have

$$\begin{aligned} (0 + 0) \cdot A &= 0 \cdot A = \emptyset = \emptyset + \emptyset = 0 \cdot A + 0 \cdot A \\ (1 + 0) \cdot A &= 1 \cdot A = A = A + \emptyset = 1 \cdot A + 0 \cdot \emptyset \\ (1 + 1) \cdot A &= 0 \cdot A = \emptyset = A + A = 1 \cdot A + 1 \cdot A. \end{aligned}$$

**6. Vector spaces.** Let  $V$  be a  $K$ -vector space and let  $V_1, V_2, V_3 \subseteq V$  be linear subspaces, none of which is contained in another. Determine with proof if  $V_1 \cup V_2 \cup V_3$  is always, sometimes or never a linear subspace of  $V$ .

*Hint:* Try different fields  $K$  to obtain examples.

*Solution:* Consider  $K^2$  and the subspaces

$$V_1 := \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle, V_2 := \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle, V_3 := \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle.$$

When  $K = \mathbb{R}$ , we have  $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in V_1$  and  $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \in V_3$ . However, their sum  $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$  is not in  $V_1 \cup V_2 \cup V_3$ . This yields an example where the union is not a linear subspace.

Now take  $K = \mathbb{F}_2$ . Then

$$V_1 \cup V_2 \cup V_3 = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}.$$

This equals the whole of  $K^2$  and therefore it is a linear subspace.

**Multiple Choice questions.** Each question can admit several answers.

**Question 1.** Which of the following sets are linear subspaces of the given vector spaces?

✓  $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid 3x_1 + 5x_2 + 3x_3 = 0, 2x_2 + x_3 = 0\} \subseteq \mathbb{R}^3$

You saw in the lectures that sets of solutions of systems of homogeneous linear equations are linear subspaces.

○  $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 3\} \subseteq \mathbb{R}^3$

It is not a linear subspace since  $(0, 0, 0)$  is not in it.

○  $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 > x_2\} \subseteq \mathbb{R}^2$

It is not a linear subspace since multiplying any element by a negative scalar  $\lambda$  reverses the inequality. Hence for any  $(x_1, x_2)$  in the set,  $\lambda(x_1, x_2)$  is not in the set anymore.

✓  $\{(0, x, 2x, 3x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^4$

It is a linear subspace since for any  $v = (0, x, 2x, 3x)$ ,  $w = (0, y, 2y, 3y)$  in the set and any  $\lambda \in \mathbb{R}$ , we have

$$v + \lambda w = (0, x + \lambda y, 2x + 2\lambda y, 3x + 3\lambda y) = (0, z, 2z, 3z),$$

for  $z = x + \lambda y$ . Hence  $v + \lambda w$  stays in the set.

○  $\{(x^4, x^3, x^2, x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^4$

It is not a linear subspace. Note that  $v := (1, 1, 1, 1)$  lies in this set but that  $2v$  does not since there is no  $x \in \mathbb{R}$  such that  $2 = x^4 = x^3 = x^2 = x$ .

**Question 2.** Consider the set of pairs of positive real numbers  $\mathbb{R}_+^2$ . The addition on  $\mathbb{R}_+^2$  is defined as follows:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} := \begin{pmatrix} x_1 y_1 \\ x_2 y_2 \end{pmatrix}.$$

We now consider three different definitions of scalar multiplication, for a  $\lambda \in \mathbb{R}$ :

- $\lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} := \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix}$
- $\lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} := \begin{pmatrix} e^\lambda x_1 \\ e^\lambda x_2 \end{pmatrix}$
- $\lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} := \begin{pmatrix} x_1^\lambda \\ x_2^\lambda \end{pmatrix}$

According to which definition of scalar multiplication does  $\mathbb{R}_+^2$  with the addition defined above become a  $\mathbb{R}$ -vector space?

- First definition

No. If  $\lambda < 0$  then  $\lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \notin \mathbb{R}_+^2$

- Second definition

No. Distributivity  $\lambda(v + w) = \lambda v + \lambda w$  is not verified.

- ✓ Third definition

It checks all of the axioms.